

Hierarichal Clustering Implementation

September 21, 2025

1 Hierarchal Clustering Implementation(Agglomerative Clustering)

```
[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import warnings
warnings.filterwarnings('ignore')
from sklearn import datasets
```

```
[4]: iris = datasets.load_iris()
iris_data = pd.DataFrame(iris.data)
```

```
[6]: iris_data.columns = iris.feature_names
```

```
[8]: iris_data.head()
```

```
[8]:      sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0                5.1             3.5             1.4             0.2
1                4.9             3.0             1.4             0.2
2                4.7             3.2             1.3             0.2
3                4.6             3.1             1.5             0.2
4                5.0             3.6             1.4             0.2
```

```
[9]: from sklearn.preprocessing import StandardScaler
scalar = StandardScaler()
X_scaled = scalar.fit_transform(iris_data)
```

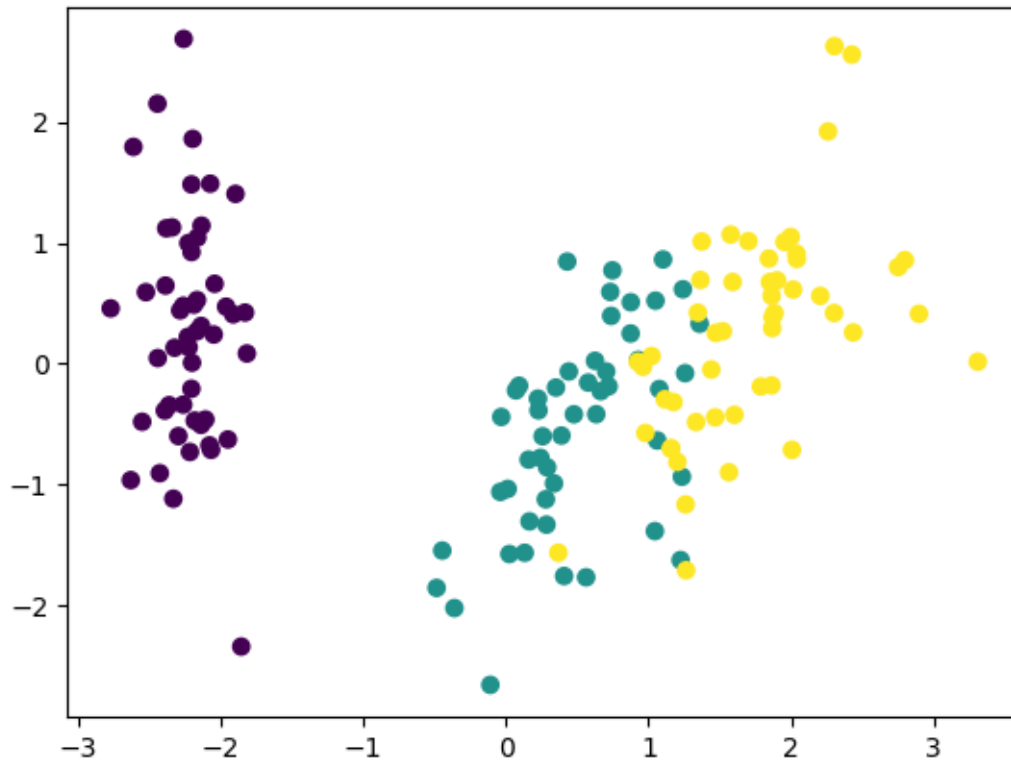
1.1 Apply PCA

```
[12]: from sklearn.decomposition import PCA
pca = PCA(n_components=2)
```

```
[13]: pca_scaled = pca.fit_transform(X_scaled)
```

```
[15]: plt.scatter(pca_scaled[:,0], pca_scaled[:,1], c = iris.target)
```

[15]: <matplotlib.collections.PathCollection at 0x2f9d3da5a80>



1.2 Agglomerative Clustering

```
[18]: ## Constructive a dendrogram
import scipy.cluster.hierarchy as sc

plt.figure(figsize=(20,20))
plt.title("Dendograms")
sc.dendrogram(sc.linkage(pca_scaled, method='ward'))
plt.xlabel("Sample Index")
plt.ylabel('Euclidian Distance')
plt.show()
```



```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0], dtype=int64)
```

1.2.1 Final Clusters

```
[23]: plt.scatter(pca_scaled[:,0], pca_scaled[:,1], c = cluster.labels_)
```

```
[23]: <matplotlib.collections.PathCollection at 0x2f9d67bf610>
```

